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(54) **TOILET TRAINING SEAT**

(71) Applicant: **MyKidzy LLC**, Terrace Park, OH (US)

(72) Inventor: **Richard M. BLOCK**, Terrace Park,
OH (US)

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(57)

ABSTRACT

A universal child's toilet trainer adaptable and usable with the conventional toilet provides an auxiliary toilet seat which in a use position lies on top of the conventional toilet seat and which in a storage position rests against the lid of the toilet seat. The toilet training seat comprises a toilet seat ring, multiple hinged wings, and a mounting plate whereby mounting plate is attached to the toilet seat lid and the hinged wings connect to the mounting plate. The toilet training seat also comprises a pick-up unit attached to or a part of the mounting plate for moving the toilet seat ring between an operational position and a storage position.





FIG. 1



FIG. 2

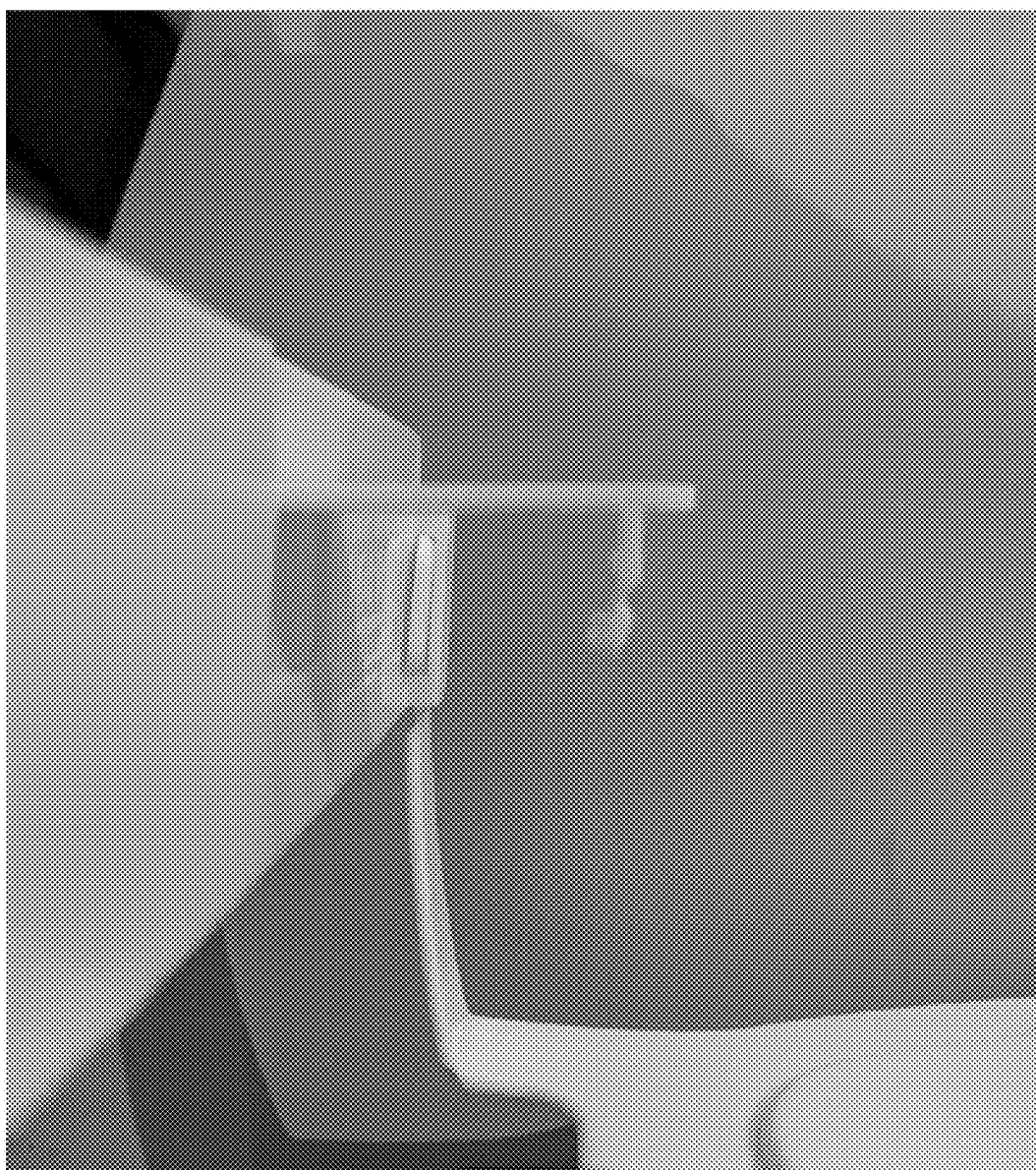


FIG. 3

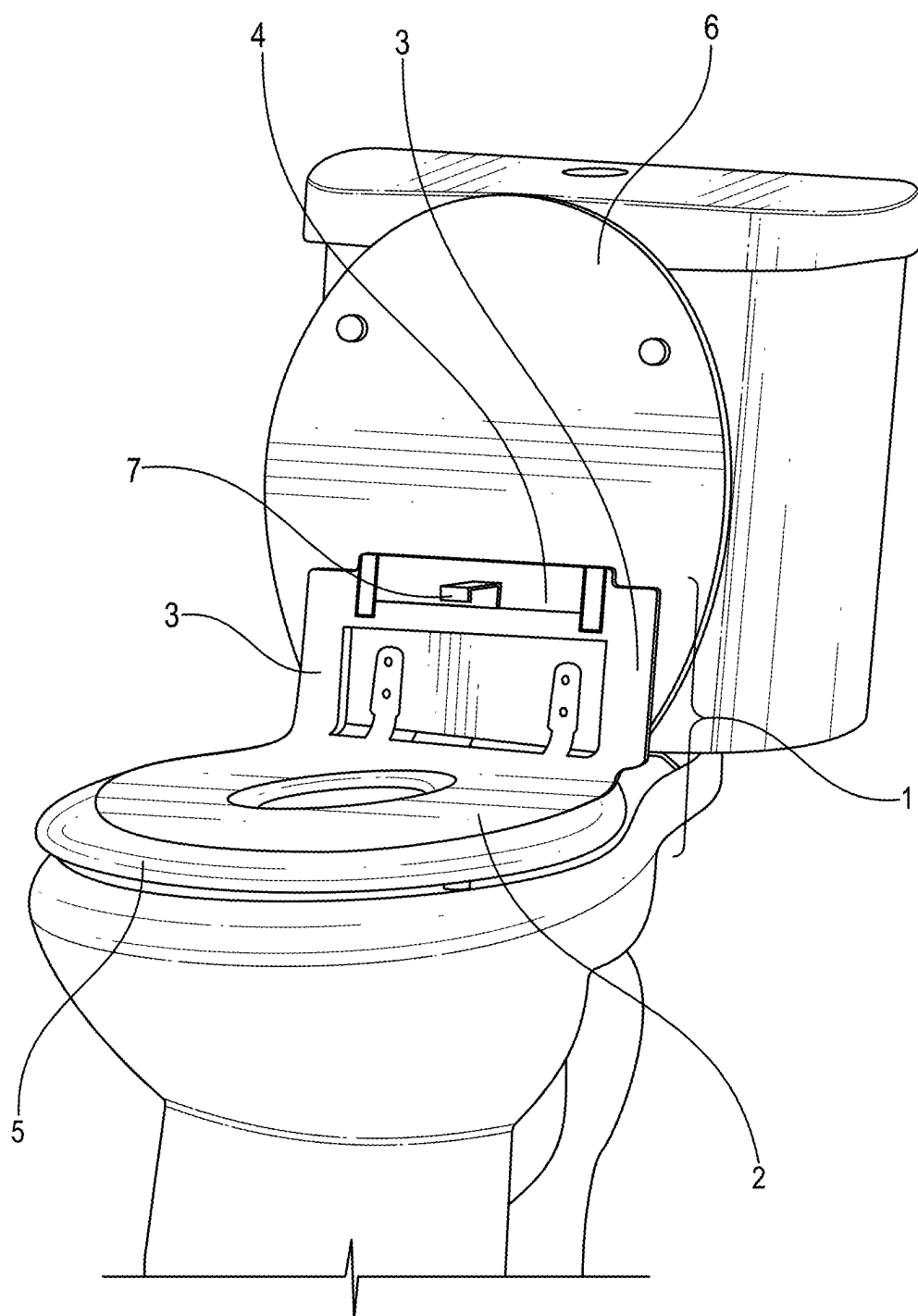


FIG. 4

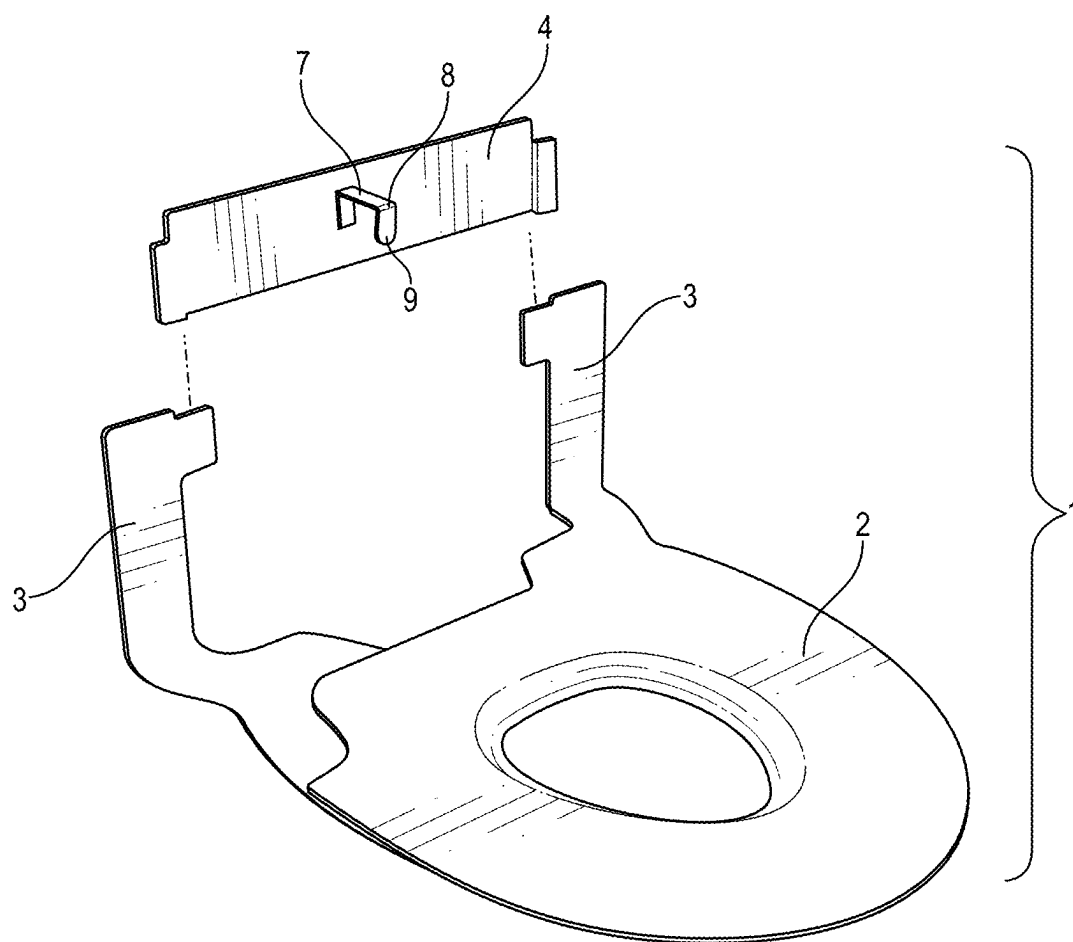


FIG. 5

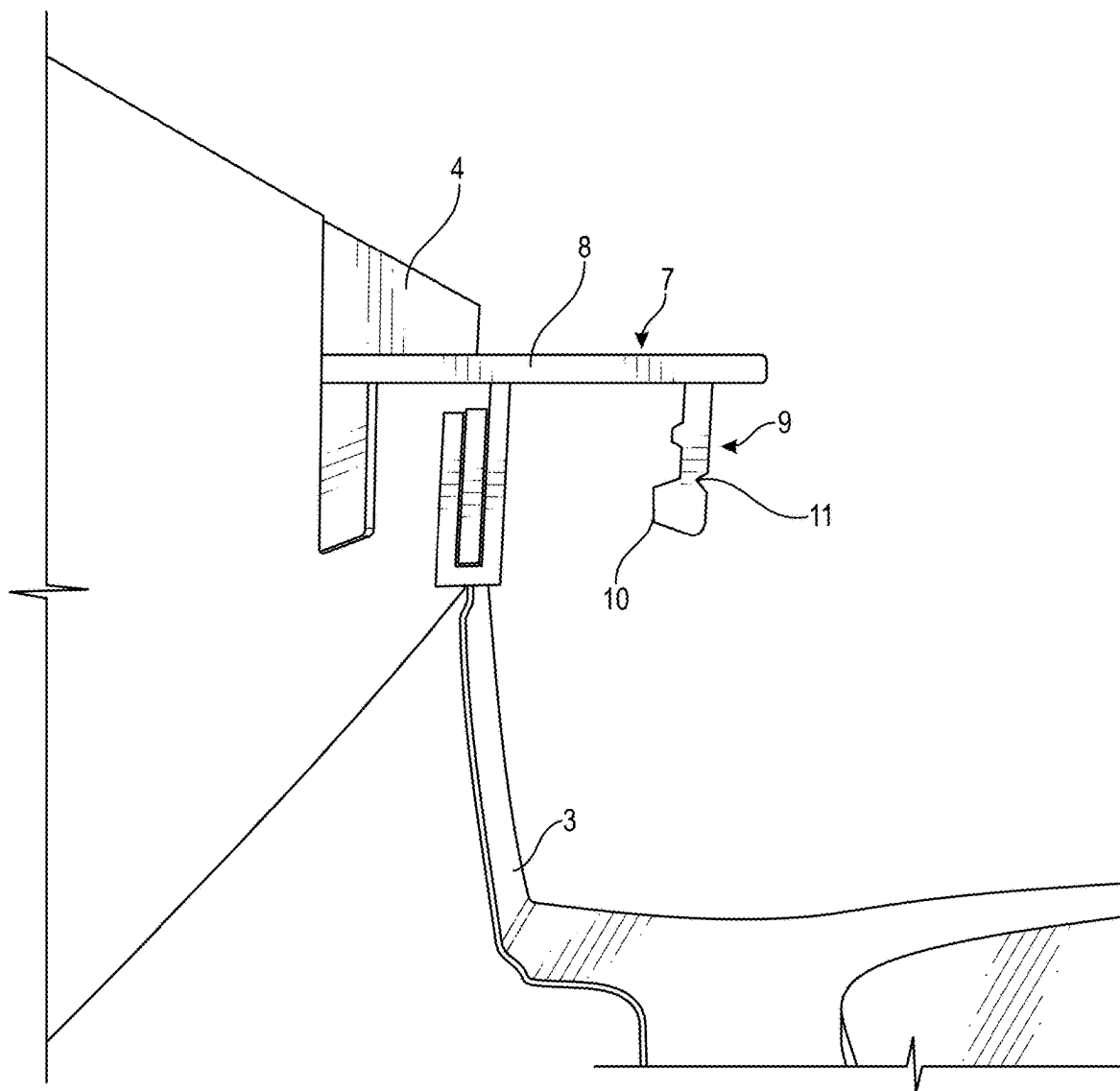


FIG. 6

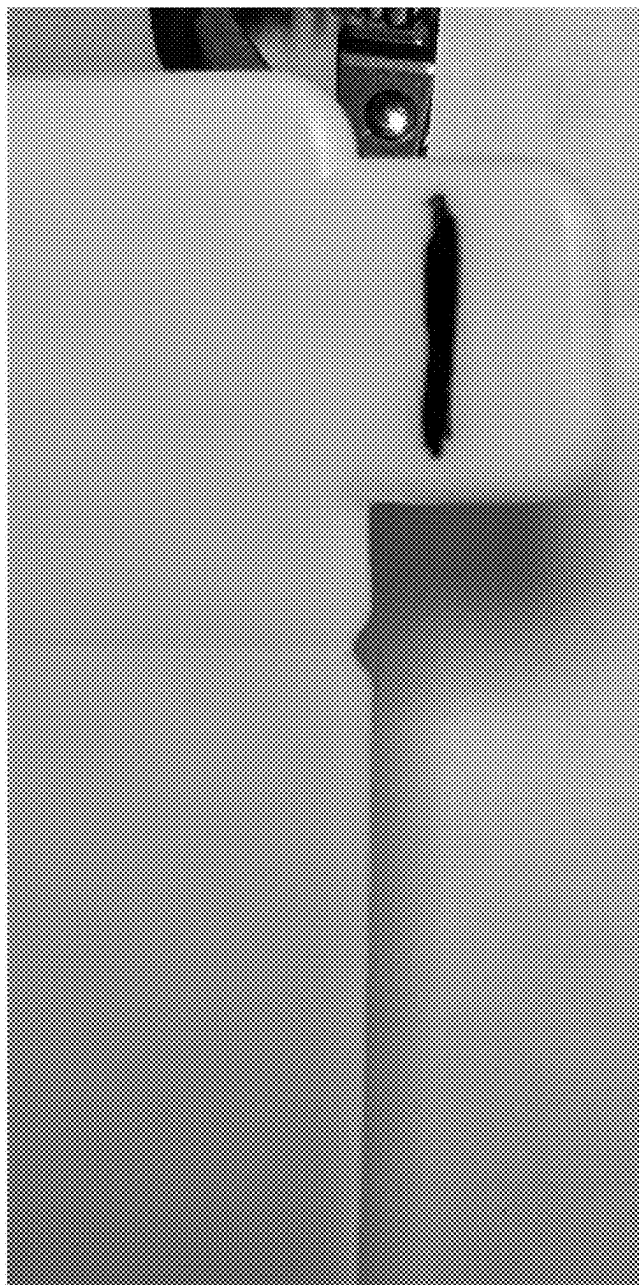


FIG. 7

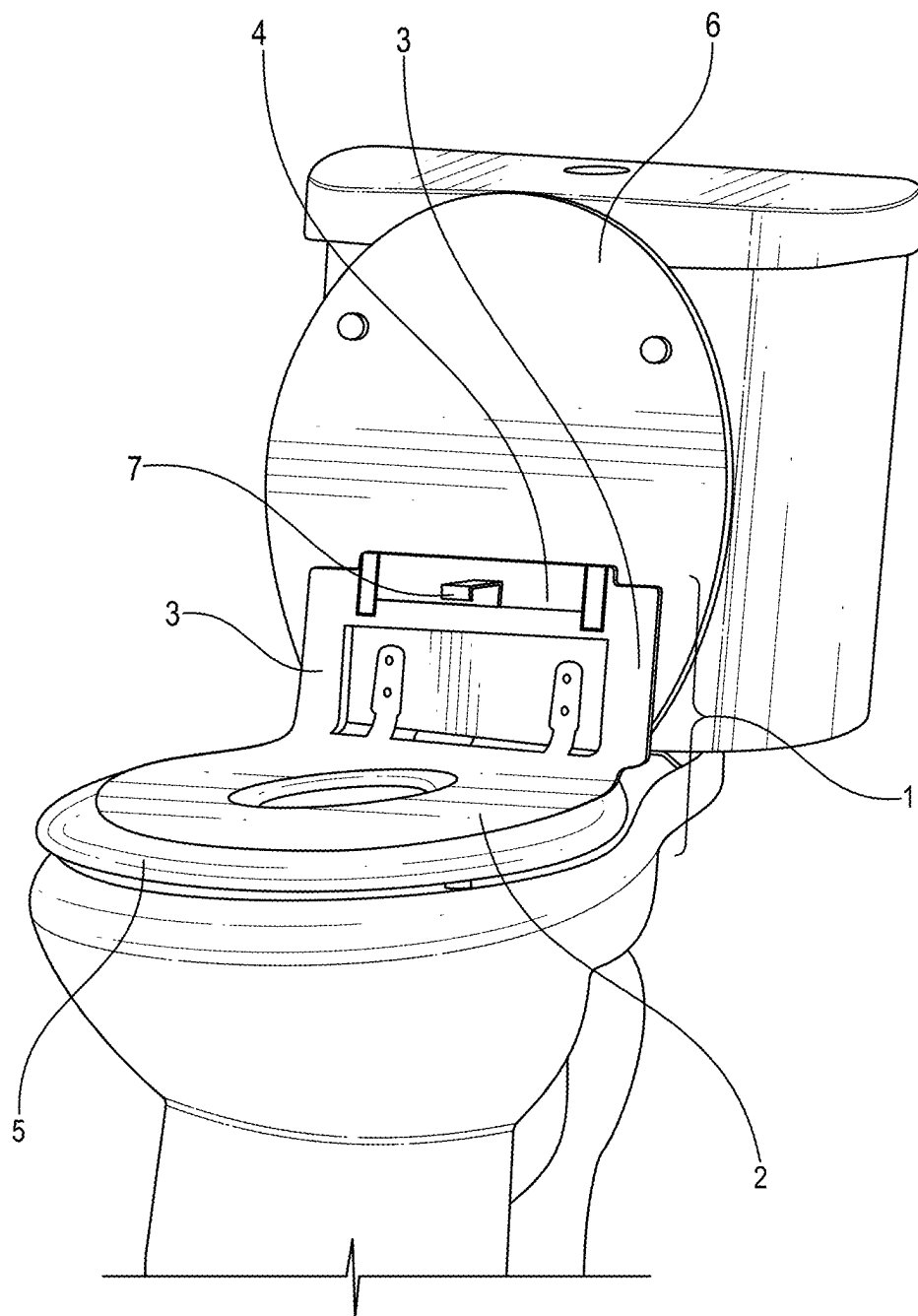


FIG. 4

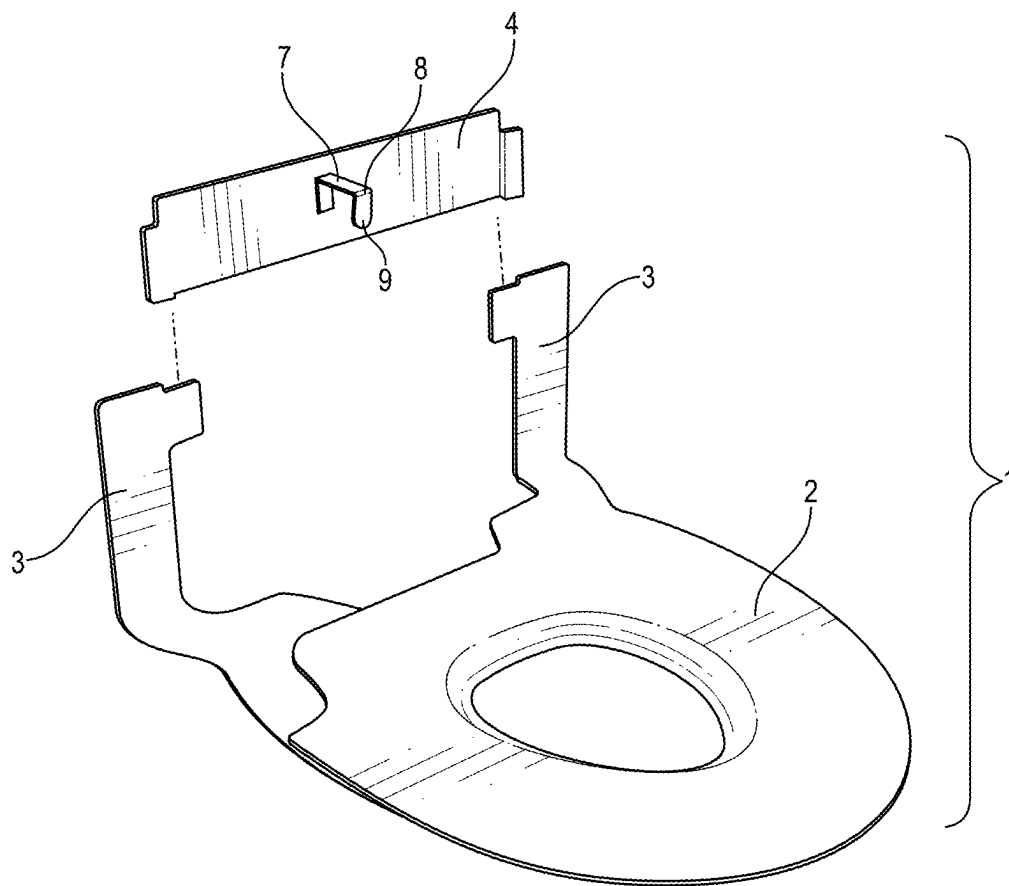


FIG. 5

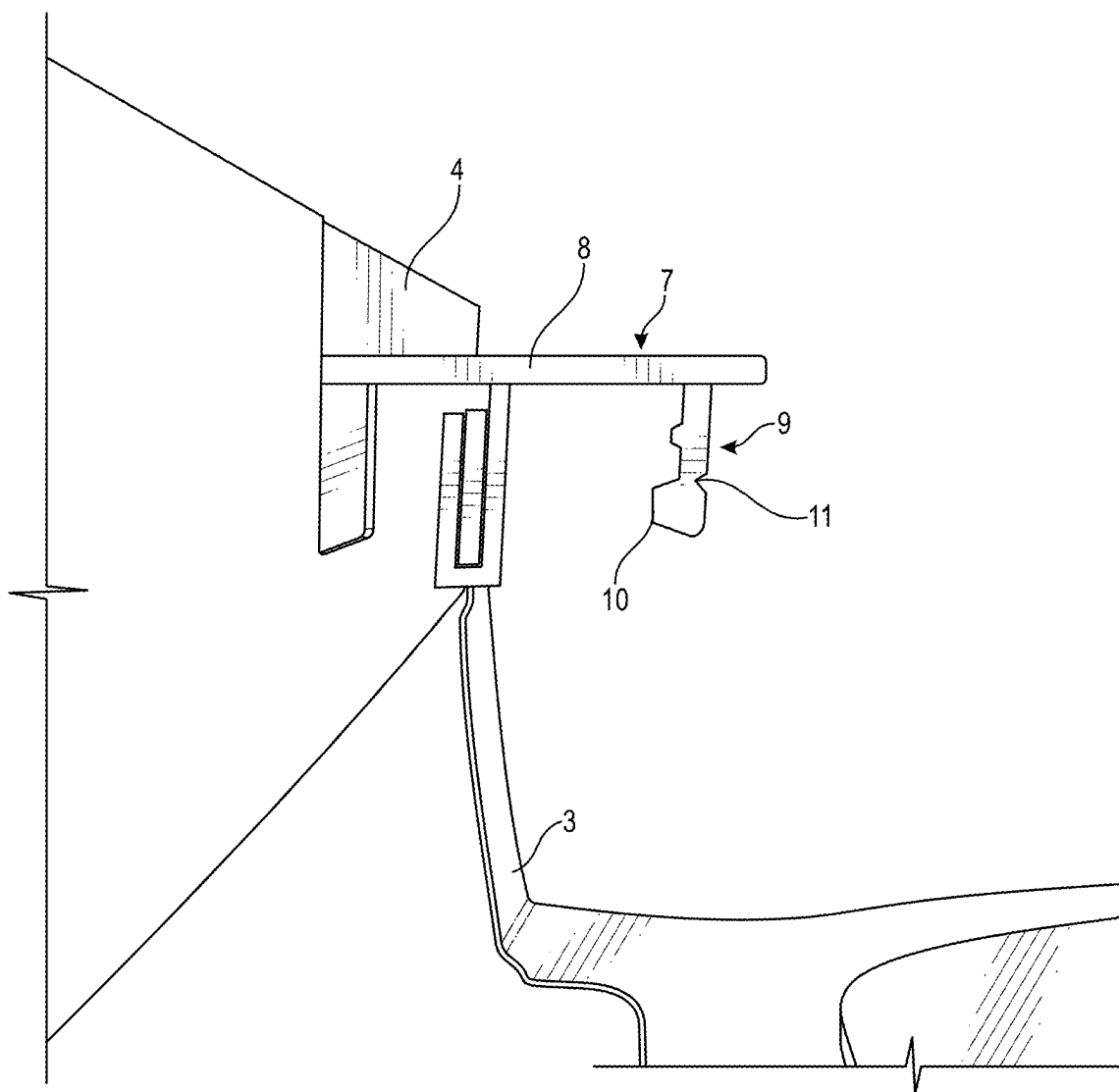


FIG. 6

TOILET TRAINING SEAT

FIELD OF THE INVENTION

[0001] The present invention relates to toilet training seats for children that is configured to be used with a conventional toilet seat and conventional toilet.

BACKGROUND

[0002] Various types of children's toilet seats and toilet trainers have been developed to provide an infant, toddler or child with an toilet seat having smaller dimensions than a conventional toilet seat. However, while meeting the problems of providing a smaller toilet seat, these prior toilet trainers create problems such as having to replace the conventional toilet seat with the training toilet seat, incapability with conventional toilet seats, sanitary issues and unattractive toilet due to the lid remaining open.

[0003] The child's toilet trainer should be usable with the conventional toilet bowl to eliminate the need of an adult handling, emptying and cleaning the toilet trainer after each use by the child and allowing for continued use the toilet by adults. Since children often require frequent use of the toilet trainer, the child's toilet seat should be capable of remaining affixed to the conventional toilet so as to always be ready to use. However, when the child's toilet seat is not being used, it should neither restrict the use of the toilet by an adult nor provide an unsightly, unaesthetic view while the child's toilet seat is stored in a nonuse position.

[0004] Prior art toilet trainers are incapable of being easily connected to the conventional toilet, but rather they require the use of special recessed lids and brackets, as well as special tools for assembly. Further, many prior art devices fail to provide a universally adaptable mechanism on a conventional toilet lid which allows the child's toilet seat to be automatically lifted from its use position to a storage position without either the need of touching the toilet seat or of using a specially designed lid.

[0005] One example of previous training seats include U.S. Pat. No. 4,516,279. However, this training seat had many additional parts, was not as secure as the training seat described herein, and had a less reliable catch or hook for raising the toilet seat ring of the training seat.

SUMMARY OF THE INVENTION

[0006] The present invention provides an improved toilet training toilet seat for children. The new seat and attaching means comes in two parts. The first part is a smaller diameter (compared to conventional toilet seats) toilet seat ring with hinged wings. The second part is a mounting plate that attaches to any standard toilet seat with double sided foam tape or other attaching means. The hinged wings are configured to slide and insert into the sides of the mounting plate, to create a singular unit.

[0007] The mounting plate has a protruding "picker-upper" latch. The protruding picker-upper latch makes contact and hooks onto the rear of the toilet seat ring. This action of the protruding picker-upper latch engaging with the smaller ring seat, attaches the smaller seat ring to the inside of the lid of any standard toilet seat.

[0008] The toilet training seat of the subject invention is then in place whenever a child needs it, yet out of the way for when an adult is using the toilet.

[0009] In embodiments of the invention, the mounting plate has an extension beyond 90 degrees of a protrusion, to create a "lift tab." This "lift tab" allows for a more sanitary way to release the toilet training seat from the latch with minimum exposure to the toilet training seat or the adult/standard toilet seat.

[0010] The picker-upper latch according to the invention includes a hinge on the underside of the lift tab. This hinge is a living hinge and is essential to proper operation of the release of the toilet training seat as it allows the toilet training seat to come down from the lid of the standard toilet seat when needed or desired.

[0011] In embodiments of the invention, the mounting plate comprises at least one insert pocket or female member that holds a male member of the hinged wings and toilet seat ring in place on the mounting plate. Each hinged wing coming off the back of the toilet seat ring has sideways extensions or male members that fit snugly into the female member insert pocket on each side of the mounting plate. The hinged wings can easily be taken in or out, on or off of the mounting plate in seconds for cleaning, travel, or storage. In some embodiments, the hinged wings have a lock knob or raised ridge on the male member. The lock knob on the hinged wings provides further safety, stability and proper operational improvement to the toilet seat trainer.

[0012] In one embodiment of the invention, the toilet training seat comprises at least one essential hinge. In specific embodiments of the invention, the toilet training seat comprises a plurality of essential hinges. In a specific embodiment, the plurality of essential hinges comprises at least 2, at least 3, at least 4, or at least 5 hinges. In a further specific embodiment, the toilet training seat comprises 4 essential hinges. In another embodiment, the toilet training seat has 5 essential hinges. This essential hinge(s) of the toilet training seat allow the seat to universally operate on all standard toilet seats.

[0013] In one embodiment of the invention, the hinge(s) of the toilet training seat are placed on the winged extensions off the back of each side of the toilet seat ring to ensure proper operation. In some embodiments, each hinged wing will have two hinges.

BRIEF DESCRIPTION OF THE DRAWINGS

[0014] The drawings are illustrative of certain embodiments of the invention and are exemplary in nature and are not intended to limit the invention defined by the claims, wherein:

[0015] FIG. 1 illustrates a front perspective view of a toilet training seat mounted on a standard toilet, according to one embodiment of the invention.

[0016] FIG. 2 illustrates a side perspective view of the toilet training seat shown in the embodiment of FIG. 1.

[0017] FIG. 3 illustrates a side view of a "lift tab" feature of the toilet training seat shown in the embodiment of FIG. 1.

[0018] FIG. 4 illustrates a front perspective view of a toilet training seat mounted on a standard toilet, according to one embodiment of the invention.

[0019] FIG. 5 illustrates side perspective view of the toilet training seat shown in the embodiment of FIG. 4.

[0020] FIG. 6 illustrates a side view of a "lift tab" feature of the toilet training seat shown in the embodiment of FIG. 4.

[0021] FIG. 7 illustrate the male member of the hinged wings with the lock knob. Also shown is one embodiment of a living hinge of the hinged wings.

DETAILED DESCRIPTION OF THE INVENTION

[0022] Specific embodiments of the invention are described herein. The invention can, however, be embodied in different forms and should not be construed as limited to the embodiments set forth herein. Rather, these embodiments are provided to illustrate more specific features of certain embodiments of the invention to those skilled in the art.

[0023] The terminology as set forth herein is for description of the embodiments only and should not be construed as limiting the disclosure as a whole. All references to singular characteristics or limitations of the present disclosure shall include the corresponding plural characteristic or limitation, and vice versa, unless otherwise specified or clearly implied to the contrary by the context in which the reference is made. Unless otherwise specified, “a,” “an,” “the,” and “at least one” are used interchangeably. Furthermore, as used in the description and the appended claims, the singular forms “a,” “an,” and “the” are inclusive of their plural forms, unless the context clearly indicates otherwise.

[0024] To the extent that the term “includes” or “including” is used in the description or the claims, it is intended to be inclusive of additional elements or steps, in a manner similar to the term “comprising” as that term is interpreted when employed as a transitional word in a claim. Furthermore, to the extent that the term “or” is employed (e.g., A or B), it is intended to mean “A or B or both.” When the “only A or B but not both” is intended, then the term “only A or B but not both” is employed. Thus, use of the term “or” herein is the inclusive, and not the exclusive use. When the term “and” as well as “or” are used together, as in “A and/or B” this indicates A or B as well as A and B.

[0025] All ranges and parameters, including but not limited to lengths, widths, percentages, concentrations, temperatures, parts, ratios and other numerical parameters disclosed herein are understood to encompass any and all sub-ranges subsumed therein, and every number between the endpoints. For example, a stated range of “1 to 10” should be considered to include any and all sub-ranges beginning with a minimum value of 1 or more and ending with a maximum value of 10 or less (e.g., 1 to 6.1, or 2.3 to 9.4), and to each integer (1, 2, 3, 4, 5, 6, 7, 8, 9, and 10) contained within the range.

[0026] As noted above, the new training seat and attaching means comes in two parts. The first part is a smaller diameter toilet seat ring 2 when compared to a conventional toilet seat. The toilet seat ring 2 includes hinged wings 3 on the rear portion of the ring. The second part is a mounting plate 4 that attaches to any standard toilet seat lid 6. The mounting plate 4 can be attached to the lid 6 with double sided foam tape, gel tape, reusable adhesive, hook and loop fasteners or other attaching means. The mounting plate can be removed from the lid of the conventional toilet seat and reused on the same lid or another toilet or toilet seat. The hinged wings are configured to slide and insert into the sides of the mounting plate, to create a singular unit 1. In some embodiments, a male member of the hinged wing slides into the female member of the mounting plate to be secured in place.

[0027] The toilet seat ring 2 has an opening or aperture, the diameter of which is smaller than the diameter of the opening or aperture in a conventional toilet seat. The opening or aperture can be about 6 to about 8 inches in length and about 4 to about 6 inches in width. In some embodiments, the opening is about 7 inches long and about 5 inches wide. In other embodiments, the opening is about 7.25 inches long by about 5.25 inches wide. The smaller dimensions are required for the toilet seat ring 2 since users of the toilet training seat, namely children, have relatively smaller body sizes compared to adults. The smaller opening prevents the child from slipping or falling during use of the toilet training seat. The toilet seat ring 2 fits directly over the conventional toilet seat opening. The toilet seat ring 2 can be made from plastic, rigid cardboard, wood, metal, a composite material or any other strong durable substance.

[0028] In an operational position as shown in FIG. 1, the toilet training seat 1 rests horizontally upon the conventional toilet seat 5. In a storage position, the toilet seat ring 2 lies against the lid 6 of the conventional toilet seat and thereby it is out of the way if an adult or child wishes to use the toilet without the toilet training seat 1. In some embodiments, the toilet seat ring 2 lies flat against the lid in the storage position. In some embodiments, the toilet seat ring 2 is parallel to the lid 6 in the storage position. The toilet seat ring 2 rests in a storage position without altering the workings of the conventional toilet seat ring 5 or lid 6. The lid 6 need not be altered or recessed to receive the toilet training seat 1, but rather the conventional lid found on any toilet can be used with the toilet training seat 1 even when the toilet training seat 1 lies upright against the lid 6 in a storage position.

[0029] A pair of hinged wings 3 connects the rear portion of the toilet training seat ring 2 to the mounting plate 4 on the lid 6 to allow movement of the toilet seat ring 2 between the storage position and the operational position. The hinged wings 3 are elongated pieces of material, preferably plastic, which can be manufactured as one integral unit with the toilet seat ring 2. This embodiment is preferred when both the toilet seat ring 2 and the hinged wings 3 are made from molded plastic. The hinged wings 3 can be made from plastic, rigid cardboard, wood, metal, a composite material or any other strong durable substance. The hinged wings extend from the rear of the toilet seat ring in one portion and also extend upward in a second portion. That is, a first portion of the hinged wings extend in the same plane as the toilet seat ring and a second portion extends at angle upward from the first portion. The second portion may extend upward at about a 90° angle from the first portion in the operational position. The angle of the first portion and second portion of the hinged wings can be any angle suitable for the described application, include about 80°, 81°, 82°, 83°, 84°, 85°, 86°, 87°, 88°, 89°, 91°, 92°, 93°, 94°, 95°, 96°, 97°, 98°, or 99°, or any other suitable angle in the operational positions. Where the first portion and the second portion of the hinged wings meet is a flexible portion that bends and flexes. This flexible portion is configured to act as a hinge. This is a living hinge. A living hinge is a thin flexible joint between two thicker and/or rigid pieces. This hinge allows the hinged wings to flex such that the toilet seat ring can be raised from the convention toilet seat and lowered from the storage position to the operational position.

[0030] In some embodiments, each hinged wing has at least one hinge. In another embodiment, each hinged wing has two hinges. In such an embodiment, the first hinge is found closer to the toilet seat ring 2 and the second hinge is found just below the male member of each hinged wing. The second hinge is shown in FIG. 7.

[0031] The mounting plate 4 is secured to the underside of the lid 6 in a position to engage the hinged wings 3. In some embodiments this position is just above the brackets which connect the lid 6 to the conventional toilet seat ring 5 and the toilet itself. The position of the mounting plate 4 on the lid 6 will vary depending the size of the toilet seat ring 2 and the opening of the toilet seat ring 2. The mounting plate 4 needs only to be sufficiently wide so as to engage or connect to the hinged wings 3. At opposite ends of the mounting plate 4 the projections or female member insert pockets extend perpendicularly outward relative to the underside of the lid 6 and, as previously noted, they receive the sideways extensions or male members of the hinged wings. In some embodiments, the male members of the hinged wings will include a lock knob or raised ridge to secure the male member in the female member such that the male member cannot easily slip out of the female member when in use. See the black line in FIG. 7, colored only to show the lock knob or ridge. While securing the male member in the female member, the lock knob also allows for the male member to be removed from the female member, if wanted or necessary. The lock knob may be positioned about 0.5 inches to about 0.75 inches from the end of the male member, preferably about 0.6 inches in from the end of the male member. The lock knob itself may be a raised ridge having a height of about 0.025 inches to about 0.075 inches and a width of about 0.075 inches to about 0.10 inches, preferably the ridge has a height of about 0.050 inches and a width of about 0.093 inches.

[0032] The mounting plate 4 can be made from plastic, rigid cardboard, wood, metal, a composite material or any other strong durable substance. Preferably, the female end insert pockets are made as one integral unit with the mounting plate 4. The backside of the mounting plate 4 has an adhesive backing which allows the mounting plate to be secured or bound to the lid 6, or alternatively, the mounting plate 4 can be attached to the lid 6 by numerous other conventional techniques, such as double sided foam tape, adhesive tapes, glues, reusable adhesives, hook and loop fasteners, screws, rivets, bolts and the like.

[0033] As shown in FIG. 3, the picker-upper latch 7 is positioned on the mounting plate 4 so that the picker-upper latch 7 engages the toilet seat ring 2 on the rear portion of the opening as lid 6 is closed down upon the toilet seat ring 2. As the lid 6 lies horizontally upon the toilet seat ring 2, the picker-upper latch 7 grasps the opening and thereby lifts the toilet seat ring 2 up from the conventional toilet seat 5 as the lid 6 is pulled up and repositioned perpendicular to the conventional toilet seat. The picker-upper latch 7 automatically, without need of a person directly touching the toilet seat ring 2, moves the toilet seat ring 2 between the operational position in which it lies horizontally on top of the conventional toilet seat 5 and the storage position in which the toilet seat ring 2 rests upright or vertically against the lid 6. When a child later wishes to use the toilet seat ring 2, the child pulls the end of the picker-upper latch away from the lid 6 or in some embodiments away from the toilet seat ring 2, and the toilet seat ring 2 is released from the picker-upper latch 7 and falls from a vertical position to a

horizontal position onto the conventional toilet seat where it again is positioned in the operational position.

[0034] Preferably, the picker-upper latch 7 is one integral unit with the mounting plate, especially if the mounting plate and picker-upper latch are made of molded plastic. The picker-upper latch can be made from plastic, rigid cardboard, wood, metal, a composite material or any other strong durable substance. Alternatively, the picker-upper latch can be a separate piece which is attached to the mounting plate by an adhesive, tape, screws or other conventional means. The picker-upper latch should be of sufficient length and width to have the requisite strength and durability to lift the toilet seat ring 2 as the lid 6 is raised from the toilet.

[0035] The picker-upper latch 7 protrudes perpendicular to the mounting plate 4. In some embodiments, the picker-upper latch protrudes at about a 90° angle from the mounting plate. The angle of the picker-upper latch 7 from the mounting plate 4 can any angle suitable for the described application, include about 80°, 81°, 82°, 83°, 84°, 85°, 86°, 87°, 88°, 89°, 91°, 92°, 93°, 94°, 95°, 96°, 97°, 98°, or 99°, or any other suitable angle. The picker-upper latch includes a plate 8 that extends or protrudes about 1 inch to about 3 inches out from the mounting plate 4. In some embodiments, the picker-upper latch includes a plate 8 that extends or protrudes about 1 inch to about 2 inches out from the mounting plate 4. In other embodiments, the picker-upper latch includes a plate 8 that extends or protrudes about 1.5 inches to about 1.75 inches out from the mounting plate 4. Plate 8 may be positioned at the top of the mounting plate 4, preferably about 0.1 inches to about 0.5 inches from the top of the mounting plate 4, more preferably about 0.25 inches from the top of the mounting plate 4. The width of the picker-upper latch plate 8 is between about 0.5 inches to about 2 inches. The picker-upper latch also includes a catch or hook 9 that is positioned toward the end of the picker-upper latch plate 8 and extends downward from the picker-upper latch plate 8. In some embodiments, the catch or hook 9 is positioned 1/8 of an inch from the end of plate 8. The catch or hook 9 is configured to engage the toilet seat ring 2. The catch or hook extends downward at about a 90° angle from the picker-upper plate 8. The angle of the catch or hook 9 from the picker-upper latch plate 8 can any angle suitable for the described application, include about 80°, 81°, 82°, 83°, 84°, 85°, 86°, 87°, 88°, 89°, 91°, 92°, 93°, 94°, 95°, 96°, 97°, 98°, or 99°, or any other suitable angle. The catch or hook 9 may be about 0.25 of an inch to about an inch in length. In some embodiments, the catch or hook 9 has a length of about 0.5 inch to about 0.75 inches. In other embodiments, the catch or hook 9 is about 0.6 inches long.

[0036] The catch or hook includes one or more nubs or ridges 10 toward the end that secures the toilet seat ring 2 when engaged. Further, the catch or hook includes a notch or hinge 11 that functions as a flex point or hinge. The notch or hinge is positioned above the nubs or ridges. This hinge 11 allows for the catch 9 to engage the toilet seat ring 2 and secure the toilet seat ring 2 to the picker-upper latch 7. This hinge 11 also allows the user to release the toilet seat ring 2 such that the toilet seat ring 2 falls into position over the conventional toilet seat in the operational position. The user will disengage the picker-upper latch 7 by pulling the end of the catch or hook away from the lid 6 such that the toilet seat ring 2 is no longer engaged with the picker-upper latch. In other embodiments, the user will disengage the picker-upper latch 7 by pulling the end of the plate 8 away from the toilet

seat ring 2 such that the toilet seat ring 2 is no longer engaged with the picker-upper latch. The hinge or notch 11 is rigid enough to retain its form and secure the toilet seat ring 2 in the up position or parallel to the lid 6 while also flexible enough to allow release of the toilet seat ring 2 by a child.

[0037] The specific embodiments and examples described herein are exemplary only and are not limiting to the invention defined in the claims. Additionally, while the present invention has been illustrated by the description of embodiments thereof, and while the embodiments have been described in considerable detail, such descriptions are not intended to restrict or in any way limit the scope of the appended claims to such detail. Additional advantages and modifications will readily appear to those skilled in the art. Therefore, the invention, in its broader aspects, is not limited to the specific details, the representative compositions and processes, or illustrative examples shown and described. Accordingly, departures may be made from such details without departing from the spirit or scope of the general inventive concept.

What is claimed is:

1. A toilet training seat comprising:
 - a first piece comprising a toilet seat ring and a plurality of hinged wing portions protruding from a portion of the ring;
 - a second piece comprising a mounting plate and a protruding latch; and
 - a lift-tab extending from a portion of the protruding latch, wherein the winged hinges comprise a male ends and the mounting plate comprises female insert pockets that correspond to and engage the male ends of the hinged wings.
2. The toilet training seat of claim 1, wherein each hinged wing further comprise a first portion that extends in the same plane as the toilet seat ring and a second portion that extends at angle upward from the first portion.

3. The toilet training seat of claim 2, wherein each hinged wing further comprise at least one flexible portion where the first portion and second portion meet wherein the flexible portion is configured to act as a hinge.

4. The toilet training seat of claim 1, wherein the mounting plate is attached to a portion of a toilet seat lid.

5. The toilet training seat of claim 4, wherein the mounting plate is attached to the toilet seat lid by one or more selected from the group consisting of an adhesive tape, a glue, a reusable adhesive, one or more screws, one or more rivets, and one or more bolts.

6. The toilet train seat of claim 1, wherein mounting plate is made from plastic, rigid cardboard, wood, metal, or any other strong durable substance.

7. The toilet training seat of claim 1, wherein the mounting plate further comprises a picker-upper latch.

8. The toilet training seat of claim 7, wherein the picker-upper latch extends perpendicular from the mounting plate.

9. The toilet training seat of claim 8, wherein the picker-upper latch comprises a plate extending perpendicular from the mounting plate and a hook that extends perpendicular downward from the plate.

10. The toilet training seat of claim 9, wherein the picker-upper latch further comprises one or more nubs or ridges positioned toward the end of the hook and a notch positioned above the nubs or ridges.

11. The toilet training seat of claim 3, wherein each hinged wing further comprise at least a first flexible portion where the first portion and second portion meet and a second flexible portion positioned below the male member of the hinged wing wherein the first flexible portion and second flexible portion are configured to act as a hinge.

12. The toilet training seat of claim 3, wherein the hinge of first flexible portion and second flexible portion is a living hinge.

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